Executive Summary — Customer Churn Prediction System

Prepared 24 October 2026 | Capstone handover

Objective

Estimate telecom customer churn risk and support prioritized, voluntary retention outreach through an end-to-end ML workflow.

Data

Model	ROC-AUC	F1	Precision	Recall
Logistic Regression	0.8447	0.6329	0.5220	0.8036
Random Forest	0.8306	0.6325	0.5359	0.7711
XGBoost	0.8426	0.6405	0.5550	0.7571

Recorded local run: 7,043 customers, 1,869 churned and 5,174 non-churned (about 26.5% churn).

Validation comparison

The recorded validation results are shown below. XGBoost had the highest F1; the validation ROC-AUC values were close.

Held-out test

ROC-AUC 0.8370; F1 0.6026; precision 0.5481; recall 0.6690. Test performance was lower than validation, so ongoing validation is necessary.

System

Data preparation, baseline/model comparison, explainability utilities, batch scoring, FastAPI, Streamlit, Docker and basic drift monitoring.

Retention hypotheses

(1) onboarding support for short-tenure/month-to-month customers; (2) plan-fit and charge review; (3) relevant support/security setup help. Test each with controlled experiments.

Limitations

Educational prototype, not production validated. SHAP describes model behavior, not causality; drift is not a performance metric. The trained model, raw data and local Docker image are generated locally and are not included in the handover ZIP.